

Social Interaction and Scientists' Productivity: Evidence from Historical Correspondence Networks in Quantum Mechanics, 1919–1932

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What is the Role of Informal Communication Among Researchers for Transmission of Knowledge and Research Productivity?

- ▶ Modern open science: online journals, ArXiv, Twitter, conferences/workshops/seminars
- ▶ Is this enough for diffusion/production of knowledge?
- ▶ Researchers discuss research ideas and comment on each other's work w/o co-authoring or citing
- ▶ Often not observable who talks to whom, and how frequently
- ▶ Common measures of social interaction in science are selected on outcomes: co-authorship, co-citation, and co-inventorship $\Rightarrow$ source of bias
- ▶ The importance of *preexisting* social ties for the transmission of knowledge, learning, and exchange of ideas

This Project: Effects of Pre-Existing Social Links to the Founders of a New Ground-Breaking Theory on the Connected Scientists' Productivity

- ▶ A major shock to science – discovery of Quantum Mechanics (Heisenberg, 1925)
- ▶ The first self-consistent framework for of the atom
- ▶ Ground-breaking nature of discovery: indeterminism (probabilistic predictions), a-causality

"I, at any rate, am convinced that He is not playing at dice." (Einstein to Born, Dec'26)

- ▶ Well-documented historical account on the development of quantum physics (manuscripts, memories, interviews, *letters*, etc.)
- ▶ Inventory of historical scientific correspondence among physicists: 5,600 physicists, 1900–1952

The Founders of Quantum Mechanics



Heisenberg



Born



Bohr



Jordan



Dirac



Pauli



Schroedinger



de Broglie

Preview of Findings: Scientists with Preexisting Social Ties to the Founders of Quantum Mechanics Produced More Scientific Output and Gathered More Citations to their Work After the Discovery (1925)

- ▶ Preexisting social ties to the founders (via correspondence) increased scientific output of preconnected scientists
- ▶ Scientists with preexisting social ties to the founders of quantum mechanics produced more scientific output and received more citations after the discovery of quantum mechanics in 1925
- ▶ Dynamic event study effects remain after filtering on the quantum studies field
- ▶ Post-1925 publications of connected scholars received more citations during the short-run after the discovery, both 5y and 25y forward citations
- ▶ Ongoing:
 - ▶ Heterogeneous effects with varying number of letters exchanged
 - ▶ Case study of selected scholars (need access to the letters) to study the potential diffusion of ideas

Physics Since World War I: A Modern Open Science

- ▶ Quick access to international academic journals (e.g., *Zeitschrift für Physik*, *Physical Review*)
- ▶ Publishing was similar to the modern process
- ▶ Conferences were very common and widely attended
- ▶ Increasing role of conferences and the telephone
- ▶ But letters (informal scientific communication) were used to discuss research and exchange ideas by the late 1960s

Selected Literature Review

- ▶ Peer effects and spillovers among researchers: Azoulay et al. (2010), Zacchia (2012), Waldinger (2012)
- ▶ Colocation of researchers and communication cost: Catalini, Fons-Rosen and Gaulé (2020), Kim et al. (2009), Agrawal and Goldfarb (2008)
- ▶ Teachers' influence in arts and sciences: Koschnick (2025), Borowiecki (2022), Tol (2022), Waldinger (2010)
- ▶ Broader effects of social interaction: diffusion of ideas in the railroad networks (Chiopris, 2023); preexisting social ties and Holocaust survival (Bělin, Jelínek & Jurajda, 2023)

Empirical evidence on the mechanisms of tacit knowledge transmission and learning among scientists is limited

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Data and Variables

- ▶ Letters: The Inventory of Sources for the History of the Twentieth-Century Physics (ISHTCP), 1900-52, 5600 physicists, all theoretical specialties [Card example](#)
- ▶ Publications and citations from the Web of Science, 1919-1932
- ▶ Dep. variable **Citation Weighted Publications** – the sum of subsequent citations (in the first 25 years after publication) to articles published in top journals by a scientist in a given year [Calculation example](#)
- ▶ Scientist-level covariates: pre-discovery productivity and quality of research; specialty; country of origin and work; birth year; year of first article.

Sample Selection and Treatment Definition

- 1 Published the first article pre-1925, stay alive by 1932
- 2 Publishing years 1919-1932
- 3 Exclude QM founders and their articles

Treatment: connection with any of the founders via correspondence (absorbing state).

Connected_i = 1 if exchanged at least one letter with any founder before 1925, 0 otherwise $\Rightarrow$ direct neighbors in the founders' ego networks

The founders of quantum mechanics

- Werner Heisenberg
- Max Born
- Niels Bohr
- Pascual Jordan
- Paul Adrian Maurice Dirac
- Wolfgang Pauli, Jr.
- Erwin Schroedinger
- Louis de Broglie

Empirical Strategy: Event Study Diff-in-Diffs with IPW/Double robust IPW (SZ, 2020; CS, 2021)

- ▶ Baseline Diff-in-Diff:

$$\log(1 + Citations_{it}) = \beta_0 + \beta_1 \times PreConnectedToFounders_i \times Post_t + \lambda_i + \lambda_t + \epsilon_{it}$$

- ▶ $\log(1 + Citations_{it})$ – Citation Weighted Publications in the first 25 yrs after publication
- ▶ $Post_t$ is 1 for years ≥ 1925
- ▶ $PreConnectedToFounders_i$ is 1 if pre-existing social link to a founder
- ▶ Scientist and Publication Year FE
- ▶ Standard errors clustered at scientist level

IPW – Reweighting the Control Group with Propensity Scores Based on Pre-Discovery Scientist-Level Covariates

- ▶ Covariates for pscore calculation (prob. of treatment for controls):
 - ▶ `prelnshare_mean`: pre-1925 mean of author share
 - ▶ `preprobp_mean`: pre-1925 mean probability of publishing
 - ▶ `prelncit3_mean`: pre-1925 mean of forward citations (next 3 years)
 - ▶ `prelncit5_mean`: pre-1925 mean of forward citations (next 5 years)
 - ▶ `i.specialty_code`: field
 - ▶ `i.country_code`: main country of work
 - ▶ `fy`: first year of publication
- ▶ Control units: $\text{weight} = \text{pscore} / (1 - \text{pscore})$
- ▶ Engineering NO pretrends in a way
- ▶ Double robust IPW method additionally subtracts the change for the control units

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After 1925, Physicists w/ Preexisting Social Ties to the Founders are **5% More Likely to Publish**, **Publish 4% More Papers**, and **Get 13% More Citations**

Period	1919–1932			
Outcome	Pr(any pub)	Nb Pubs	Cites 5y	Cites 25y
	(1)	(2)	(3)	(4)
$PreConnected_i \times Post_t$	0.054** (0.022)	0.042*** (0.024)	0.123*** (0.041)	0.131*** (0.052)
Ind. and Year FE	Yes	Yes	Yes	Yes
Mean outcome treated	0.428	0.441	0.475	0.668
Nb of clusters	1,343	1,343	1,343	1,343
Obs	18,802	18,802	18,802	18,802
R ²	0.239	0.315	0.337	0.325
Adj R ²	0.180	0.262	0.286	0.273

Notes: Dep. variables are the $\log(1 + x)$, except for Pr(any pub). Standard errors are clustered at the scientist level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Pre-1925 Balancing

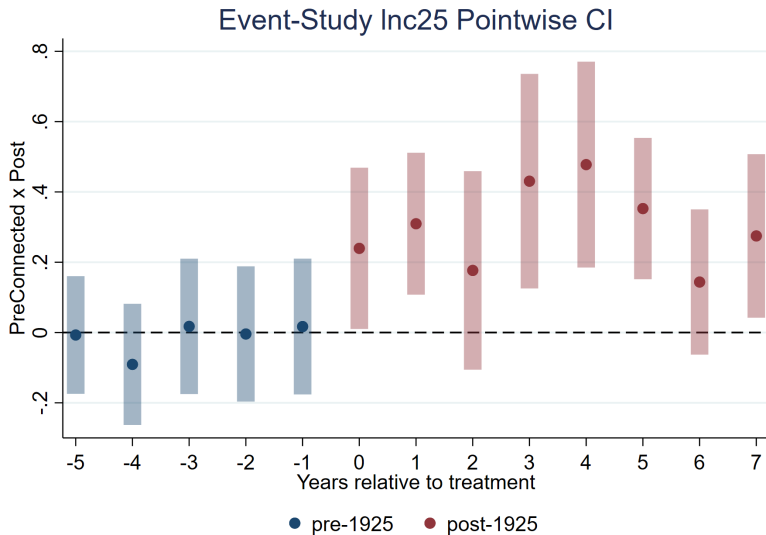
	Treated	Control
Pr(any pub)	0.428 (0.495)	0.295 (0.456)
Nb Pubs $[\ln(1+x)]$	0.441 (0.575)	0.259 (0.433)
Citations, 5y $[\ln(1+x)]$	0.475 (0.879)	0.181 (0.514)
Citations, 25y $[\ln(1+x)]$	0.668 (1.116)	0.300 (0.739)
Citations, 50y $[\ln(1+x)]$	0.769 (1.257)	0.363 (0.865)
First Pub Year	1921.14 (2.076)	1921.57 (2.020)
Observations	1,812	6,246

Event Study (CS 2021): the event-study coefficients are ATTs at each event time

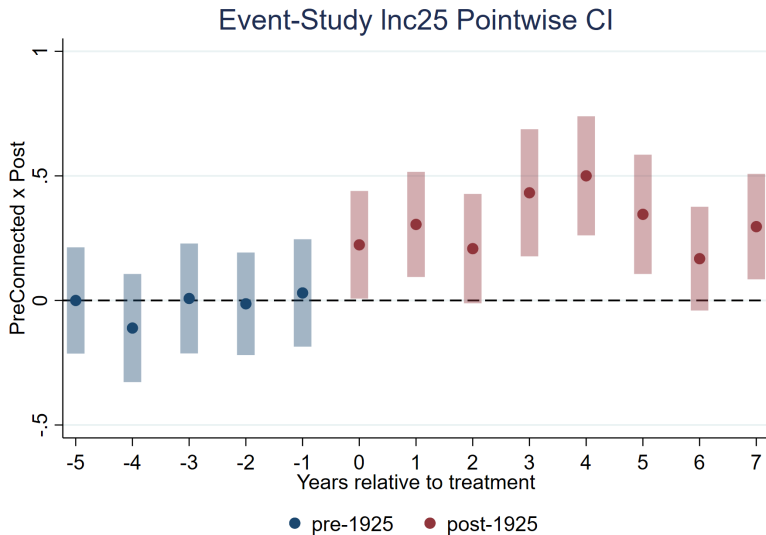
$$y_{it} = \alpha_i + \lambda_t + \sum_e \beta_e \mathbf{1}\{t - 1925 = e\} \times PreConnected_i + \varepsilon_{it}$$

- ▶ 1925 is the treatment year of scientist i .
- ▶ $e = t - 1925$ is event time.
- ▶ β_e measures the event-time ATT at each event time.
- ▶ In my setting, treatment, or the shock (the discovery of quantum mechanics) happens at the same time for all pre-connected scientists.
- ▶ y_{it} measures changes in publication outcomes:
 - ① Indicator equal to 1 if scholar i publishes at least once in year t
 - ② Number of publications by scholar i in year t

Event Study Diff-in-Diff with IPW: $\text{Log}(1 + \text{Citations } 25\text{y})$



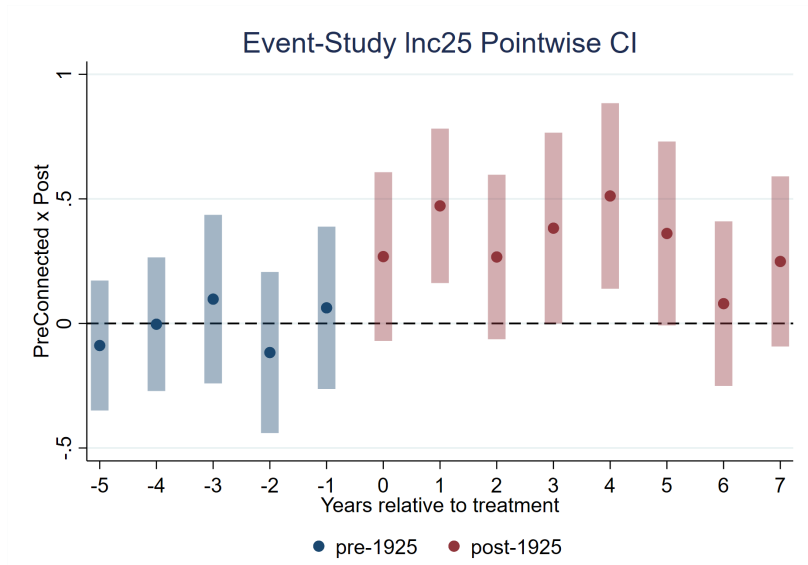
Doubly Robust IPW Event Study Diff-in-Diff: $\text{Log}(1 + \text{Citations } 25\text{y})$



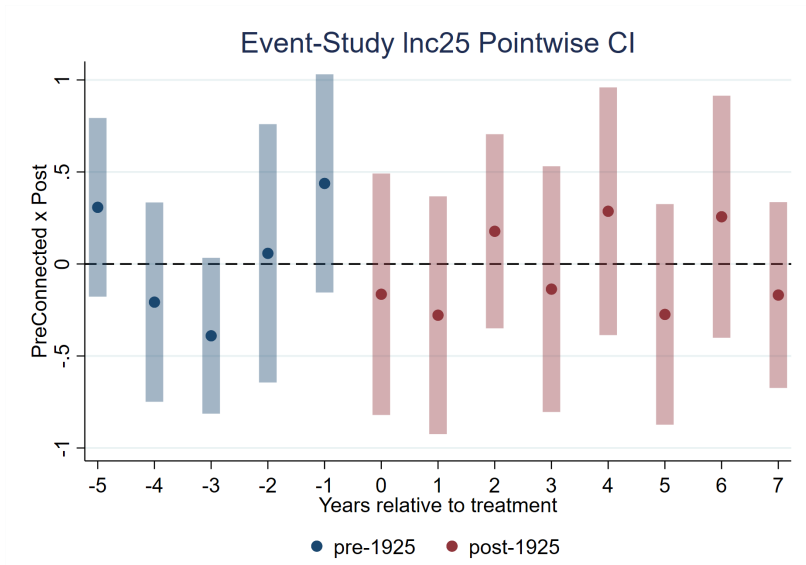
Some Robustness Check: Filter on a Narrow Group (Quantum theoretical studies)

- ▶ 8,008 obs left (out of 18,802)
- ▶ 572 clusters, 392 control, 180 treated
- ▶ Baseline effect on $\log(1 + \text{Cites } 25\text{y})$ is $\approx 13\%$

Doubly Robust IPW Event Study: $\text{Log}(1 + \text{Citations } 25\text{y})$, Only Theoretical Quantum Studies



Doubly Robust IPW Event Study: $\text{Log}(1 + \text{Citations } 25\text{y})$, Astrophysics and Physical Chemistry



Conclusion

- ▶ I study whether the *preexisting* social ties matter for the transmission of knowledge, learning, and exchange of ideas
- ▶ I focus on a major break-through event in the twentieth-century physics - quantum mechanics (1925)
- ▶ I approximate social ties with the presence and intensity of scientific correspondence of a scientist to the founders of quantum mechanics
- ▶ Scientists with preexisting social ties to the founders of quantum mechanics produced more scientific output and received more citations after the discovery of quantum mechanics in 1925
- ▶ Contribution to the literature on communication cost, collaboration, peer effects and the diffusion of knowledge

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Example of citations calculation

If a researcher published two top-journal articles in 1932, cited 5 and 7 times within 50 years, the citation-weighted publication measure for 1932 is $(5 + 7 = 12)$.

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Example #1 of a microfilm card

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INVENTORY OF SOURCES FOR HISTORY OF TWENTIETH-CENTURY PHYSICS -- (ISHCP)												
P	C	Archive	Years: P > C	exchg	C > P	Years: P > C	exchg	C > P	Years: P > C	exchg	C > P	
<i>Eve, A/Wilson, H</i>												
		Slater, John Clarke US.Ph.01.b	---		46-52:	4/	4	/	3/	4	* 52-++:	/
		Eversheim, Paul			(1873-			)				
		Darmstaedter, Ludwig GW.Be.01.g	---		22-25:	1/		/				
		Evjen, H. M.			(	-		)				
		Ewald, Heinz			(1914-			)				
		Ewald, Peter Paul			(1888-			)				
		Bernal, John Desmond GB.Ca.01.e GB.Ca.01.e	---		39-45:	1/	2	/	1/	1		
		Bethe, Hans Albrecht US.It.01.a	---		39-45:	3/	3	/	3/	3		
		Born, Max GW.Be.01.e	53-++:	1/	1	/			32-38:	1/	2	
		See: 3/5-9			nd:	1/	3	/	/			
		Bothe, Walther Wilhelm Georg GW.Be.03.c	---		32-38:	1/	1	/	/			
		Bragg, William Henry GB.Lo.03.a GB.Lo.03.a	---		14-21:	1/	4	/	/			
			---		14-21:	1/	4	/	/			
			32-38:	1/	2	/			* 22-25:	1/	1	
									* 22-25:	1/	1	
										3/	3	
P	C	Archive	Years: P > C	exchg	C > P	Years: P > C	exchg	C > P	Years: P > C	exchg	C > P	
x/y = x letters on y pages; * = unknown number of letters; P > C = physicist (or C > P = correspondent) is writer, or exchanged. page 3848												

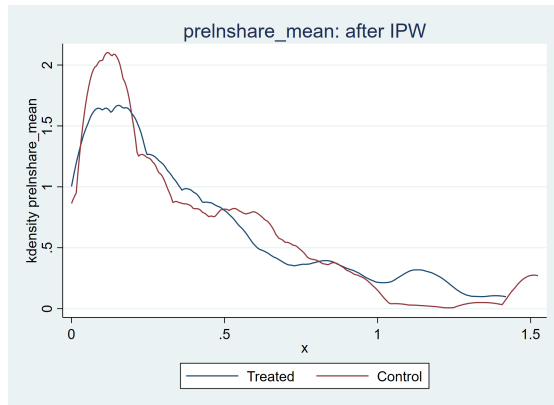
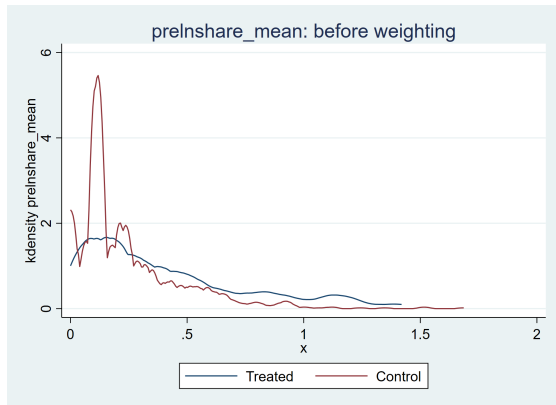
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Example #2 of a microfilm card

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INVENTORY OF SOURCES FOR HISTORY OF TWENTIETH-CENTURY PHYSICS -- (ISHICP)													
P	C	Archive	Years:	P > C	exchg	C > P	Years:	P > C	exchg	C > P	Years:	P > C	exchg C > P
<i>Bohr, N/Marke, A</i>													
		Marke, A. W. Dk.Co.01.a	Mf: SC 5.1		----	14-21:	1/ 4 /		1/ 4				
		Marsden, Ernest Dk.Co.01.a	Mf: SC 5.1; 29.4 46-52: /	/	----	07-13: 2/ 8 / 53-54: 6/ 6 /		1/ 4 * 14-21: 1/ 4 /		9/ 13			
		Marshall, George Dk.Co.01.b			----	46-52: 1/ /							
		Marvin, C. H. US.Wa.11.a			----	32-38: 1/ 2 /		2/ 2					
		Mayer, Maria Goeppert US.LJ.01.a US.NY.02.0			----	53-54: / / 46-52: 1/ 1 /		1/ 1					
		McLaren, S. B. Dk.Co.01.a	Mf: SC 5.1		----	07-13: 2/ 7 /		2/ 4					
		McLennan, John Cunningham Dk.Co.01.a	Mf: SC 5.1; 14.1		----	14-21: 2/ 3 /		2/ 4 * 22-25: 10/ 14 /		16/ 25			
		Mecke, Reinhold Dk.Co.01.b			----	14-21: / /		1/ 1					
		Meggers, William Frederick Dk.Co.01.a US.NY.02.c	Mf: SC 14.1		----	22-25: 2/ 3 / 22-25: 2/ 3 /		2/ 3 * 26-31: / /		4/			
		Melner, A. Dk.Co.01.a	Mf: SC 31.2		----	46-52: 1/ 1 /		1/ 1					
		Meltner, Lise Dk.Co.01.a	Mf: 23; 29 53-54: 1/ 1 /	/	----	22-25: / /		1/ 2 * 46-52: 1/ 1 /		1/ 2			
		GB.Ca.03.d	32-38: 1/ 1 /	/	----	2/ 3 * 30-45: 5/ 6 / 53-54: / /		1/ 1 * 26-31: 1/ 1 /		2/ 4			
		GB.Ca.03.d			----			4/ 5 * 46-52: 1/ 1 /		1/ 1			
P	C	Archive	Years:	P > C	exchg	C > P	Years:	P > C	exchg	C > P	Years:	P > C	exchg C > P
x/y = x letters on y pages; * = unknown number of letters; P > C = physicist (for C > P = correspondent) is writer, or exchanged. page 913													

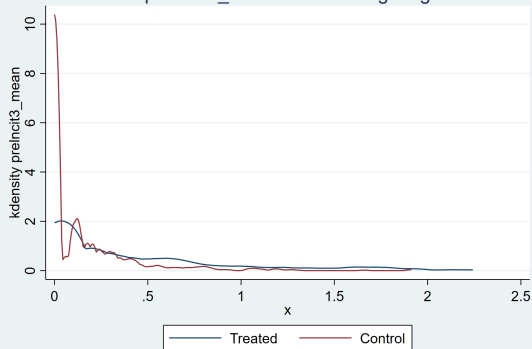
Distribution of pre-1925 Share of authors: before and after reweighting



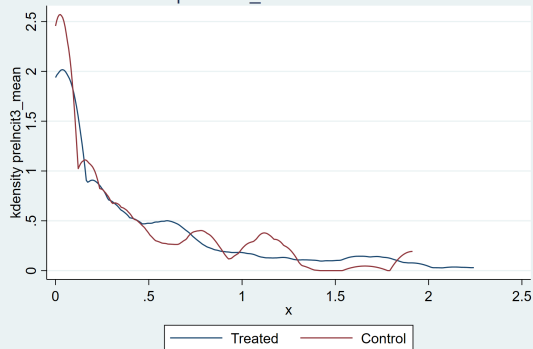
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Distribution of pre-1925 Cites 3y: before and after reweighting

preIncit3_mean: before weighting



preIncit3_mean: after IPW



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No-anticipation assumption

- ▶ Started off by Heisenberg (1925) and Schrödinger (1926) *independently* from each other
 - ▶ Unexpected *simultaneous* discovery
 - ▶ *Unprecedentedly fast* path of discovery – basic principles completed by 1927
- ▶ Two different theories about the same nature phenomena
 - ▶ Heisenberg's Matrix mechanics: anti-causal, probabilistic, discrete and with new formalism (infinite matrices)
 - ▶ Schrödinger's Wave mechanics: causal, deterministic, continuous and with old formalism (differential equations)
- ▶ Immediately accepted
 - ▶ Matrix Mechanics $\Rightarrow$ Göttingen-Munich-Copenhagen
 - ▶ The matrix mechanics, by the end of 1925, was recognized as an important advance throughout the German-speaking Central-European physicists and abroad, where was good relation with this Göttingen: Leiden, Cambridge-Engl., Cambridge-Mass., Pasadena (California)
 - ▶ Wave Mechanics $\Rightarrow$ other German-speaking physicists and countries outside of Europe
 - ▶ However, Schroedinger's theory was accepted enthusiastically among causalists. Outside of Germany it was adopted far more widely and enthusiastically than was the matrix mechanics.